

Supercomputing In India

The PARAM 8000 was the first step in India's quest for high performance computing. The 8000 was followed by the Param 9000 and Param 10000, in 1996 and 1998 respectively, with speeds of 2 Gigaflops per second and 100 Gigaflops per second respectively, using the Sun Sparc and Sun UltraSparcII microprocessors, respectively. Most C-DAC supercomputers have been used for managing the environmental data received from satellites, though it is believed that simulations of the second nuclear explosion in 1998 were carried out as well (which led to its blacklisting, along with 98 other organisations by the US government). Supercomputers have been created by other organisations in the country too. The BARC once created a supercomputer for its use, in 1991, called Anupam 860/4, using an Intel 860 microprocessor. Other more powerful versions have followed (Warning : Acronym Overload!) The ANURAG (Advanced Numeric Research and Analysis Group) of the DRDO (Defence Research and Development Organisation) developed the PACE (Processor for Aerodynamic Computations and Evaluation) supercomputer, capable of 8 Gigaflops per second, in 1995, to simulate air flow patterns for Light Combat Aircraft. The PACE PLUS 32, a more powerful version of the PACE, capable of 30 Gigaflops per second, came in 1998, and is used for missile development. HAL (Hindustan Aeronautics Limited) created a supercomputer, the Supersolver (in collaboration with Tata-Elxsi), which uses SGI R8000 CPUs, and is capable of 1.2 Gigaflops.

July: India's First Supercomputer

India's first supercomputer, Param 8000, was created. After the US government denied the export of a Cray supercomputer to India in the 1980s; the C-DAC (Centre for Development of Advanced Computing) was created in 1988 for the purpose of building an indigenous supercomputer. In the span of three years, the Param 8000, based on the massively-parallel processing architecture, had been completed. It was twice as fast as a "serial" Cray of that time—and cost about a third. Incorporating 256 nodes, each containing